

CHAPTER - II

Change of Axes

2.1 Transformation of co-ordinates

The co-ordinates of a point depend upon the origin and the axes of co-ordinates chosen. The choice of the position of co-ordinate axes is quite arbitrary. The co-ordinate of a point with references to one set of axes will change as soon as a new set of axes are chosen.

The equation of a curve will also be changed by such transformation. In the discussion of problems it is very often found advantageous to alter the origin without alteration of the direction of axes, or by altering the direction of the axes and keeping the origin unchanged, or by altering the origin and also the directions of axes. The latter case is merely a combination of the first two. Either of these processes is called *a transformation of co-ordinates*. We proceed to establish the fundamental formulae for such transformation of co-ordinates.

2.2 Translation of axes (change of origin)

To find the change in the co-ordinates of a point when the origin is shifted to another point but the direction of axes remain unaltered.

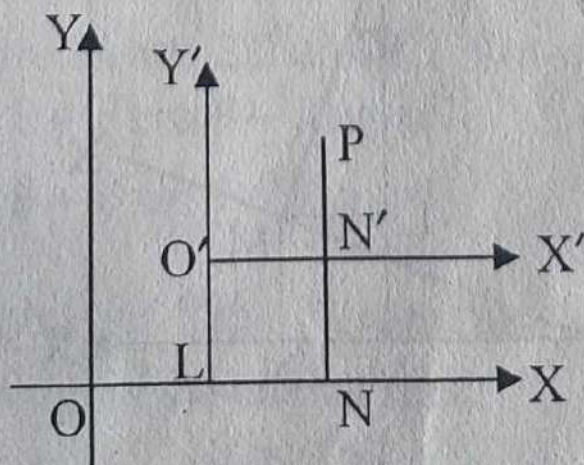


Fig.2.1

Let OX and OY be the original axes and let the new axes parallel to the original, be $O'X'$ and $O'Y'$.

Let the co-ordinates of the new origin O' , referred to the original axes be h and k , so that, if $O'L$ be perpendicular to OX , we have $OL = h$ and $LO' = k$. Let P be any point in the plane having co-ordinates (x, y) referred to the original axes and (x', y') referred to the new axes. Draw PN perpendicular to OX to meet $O'X'$ at N' . Then $ON = x$, $NP = y$, $O'N' = x'$ and $N'P = y'$.

$$\therefore x = ON = OL + LN = OL + O'N' = h + x'$$

$$\text{and } y = NP = NN' + N'P = LO' + N'P = k + y'$$

Thus, to transfer the origin to the point (h, k) , the formulae of transformation are $x = x' + h$ and $y = y' + k$.

It is to be noted that when the origin is transferred to the point (h, k) , the transformed equation of a curve is obtained by substituting $x + h$ and $y + k$ for x and y respectively in the equation of the curve.

2.3 Rotation of axes (Origin fixed)

To find the change in the co-ordinates of a point when the directions of axes is turned through an angle but the origin of co-ordinates remains the same.

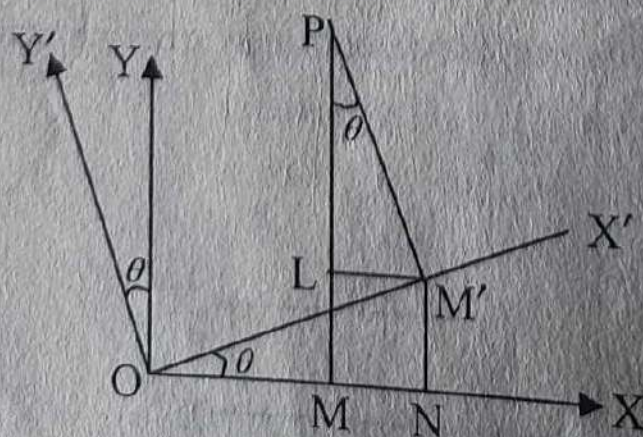


Fig.2.2

Let OX, OY be the original system of axes and OX', OY' be the new system through the same origin O and let θ be the angle through which the axes are turned, so that $\angle XOX' = \angle YOY' = \theta$. Let the co-ordinates of any arbitrary point P referred to the original axes be (x, y) and let these referred to the new axes be (x', y') . Draw PM, PM' perpendicular to OX, OX' and $M'N$ perpendicular to OX and $M'L$ perpendicular to PM . Then, $OM = x, MP = y, OM' = x', PM' = y'$ and

$$\angle LPM' = 90^\circ - \angle PM'L = \angle LM'O = \angle XOX' = \theta.$$

$$\begin{aligned} \therefore x = OM &= ON - MN = ON - LM' \\ &= OM' \cos \theta - PM' \sin \theta = x' \cos \theta - y' \sin \theta \end{aligned}$$

$$\begin{aligned} \text{and } y = PM &= ML + LP = NM' + LP = OM' \sin \theta + PM' \cos \theta \\ &= x' \sin \theta + y' \cos \theta \end{aligned}$$

Thus, to turn the axes through an angle θ , the formulae of transformation are

$$x = x' \cos \theta - y' \sin \theta \quad \text{and} \quad y = x' \sin \theta + y' \cos \theta$$

It is to be noted that when the direction of axes are turned through an angle θ , the transformed equation of a curve is obtained by substituting $(x \cos \theta - y \sin \theta)$ and $(x \sin \theta + y \cos \theta)$ for x and y respectively in the equation of the curve.

Observation: When it is required to change the origin to (h, k) as well as to turn the directions of axes through an angle θ , the formulae of transformation will obviously be the combination of the formulae obtained in arts 2.2 and 2.3, that is

$$\left. \begin{aligned} x &= h + x' \cos \theta - y' \sin \theta \\ y &= k + x' \sin \theta + y' \cos \theta \end{aligned} \right\}$$

2.4 Simplification of the equation of a curve by transformation of co-ordinates:

By proper choice of co-ordinate axes, the equation of any curve may be simplified to a great extent. This is effected in two

different ways (i) by removing the terms of the first degree in x and y and (ii) by removing the product term in x and y of an equation. Let us consider the general equation of 2nd degree in x and y i.e., $ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$. Let us shift the origin to the point (α, β) , then the new equation will be

$$a(x' + \alpha)^2 + b(y' + \beta)^2 + 2h(x' + \alpha)(y' + \beta) + 2g(x' + \alpha) + 2f(y' + \beta) + c = 0$$

$$\text{i.e., } ax'^2 + by'^2 + 2hx'y' + 2x'(a\alpha + h\beta + g) + 2y'(b\beta + h\alpha + f) + (a\alpha^2 + b\beta^2 + 2h\alpha\beta + 2g\alpha + 2f\beta + c) = 0.$$

Now, suppose we want to remove the first degree terms of the equation, then α and β are so chosen that the coefficients of x' and y' should vanish. Hence equating the coefficients of x' and y' to zero, we get $a\alpha + h\beta + g = 0$ and $b\beta + h\alpha + f = 0$.

Solving these two equations, we get

$$\alpha = \frac{bg - fh}{h^2 - ab} \text{ and } \beta = \frac{af - gh}{h^2 - ab}$$

Hence, we can conclude that the co-efficients of the first degree terms of an equation of second degree may be vanished by shifting the origin to the point $\left(\frac{bg - fh}{h^2 - ab}, \frac{af - gh}{h^2 - ab} \right)$, provided $h^2 \neq ab$. If

$h^2 = ab$, we cannot get any finite value of α and β . The quantity $h^2 - ab$ of a general equation of second degree is known as *discriminant* of the equation.

It is also observed from the new equation that the coefficients of the second degree terms remain unaltered by this transformation.

Let us now consider the expression $ax^2 + 2hxy + by^2$ and let us rotate the axes through an angle θ about the origin. To get the

value of the above in terms of new co-ordinates, we substitute $x' \cos \theta - y' \sin \theta$ and $x' \sin \theta + y' \cos \theta$ for x and y respectively in the expression so that it becomes

$$a(x' \cos \theta - y' \sin \theta)^2 + 2h(x' \cos \theta - y' \sin \theta)(x' \sin \theta + y' \cos \theta) + b(x' \sin \theta + y' \cos \theta)^2.$$

On simplification, it becomes

$$\begin{aligned} & (a \cos^2 \theta + b \sin^2 \theta + 2h \sin \theta \cos \theta)(x')^2 \\ & + (a \sin^2 \theta + b \cos^2 \theta - 2h \sin \theta \cos \theta)(y')^2 \\ & + \{2h(\cos^2 \theta - \sin^2 \theta) - 2(a - b) \sin \theta \cos \theta\} x' y' \end{aligned}$$

Now suppose we want to remove the product term i.e., $x'y'$ -term, the coefficients of $x'y'$ should be zero and hence we get

$$2h(\cos^2 \theta - \sin^2 \theta) - 2(a - b) \sin \theta \cos \theta = 0,$$

$$\text{i.e., } 2h \cos 2\theta - (a - b) \sin 2\theta = 0$$

$$\text{i.e., } \tan 2\theta = \frac{2h}{a - b} \quad \text{i.e., } \theta = \frac{1}{2} \tan^{-1} \left(\frac{2h}{a - b} \right)$$

So if it is necessary to remove the product term of an equation, the axes should be rotated through an angle θ such that

$$\theta = \frac{1}{2} \tan^{-1} \left(\frac{2h}{a - b} \right).$$

2.5 Invariants

If due to transformation of axes, the value of a particular quantity of the original equation remains unaltered in the new equation then that quantity is called an *invariant* of the equation with respect to the transformation.

We have seen already that due to transformation the coefficients of x^2 , y^2 and xy of a 2nd degree equation remain unaltered. So these coefficients of a second degree equation are invariants due to translation of axes.

Let us now consider rotation of axes through an angle θ and let the expression $ax^2 + 2hxy + by^2$ becomes $a'x'^2 + 2h'xy' + b'y'^2$ by this transformation so that

$$a' = a \cos^2 \theta + b \sin^2 \theta + 2h \sin \theta \cos \theta$$

$$b' = a \sin^2 \theta + b \cos^2 \theta - 2h \sin \theta \cos \theta,$$

$$\text{and } 2h' = 2h(\cos^2 \theta - \sin^2 \theta) - 2(a - b) \sin \theta \cos \theta, \text{ [by art 2.4]}$$

$$\begin{aligned} \text{Now, } a' + b' &= a \cos^2 \theta + b \sin^2 \theta + 2h \sin \theta \cos \theta + a \sin^2 \theta \\ &\quad + b \cos^2 \theta - 2h \sin \theta \cos \theta \\ &= a(\cos^2 \theta + \sin^2 \theta) + b(\sin^2 \theta + \cos^2 \theta) \\ &= a + b. \end{aligned}$$

$$\begin{aligned} \text{and } a'b' - (h')^2 &= (a \cos^2 \theta + b \sin^2 \theta + 2h \sin \theta \cos \theta) \\ &\quad \times (a \sin^2 \theta + b \cos^2 \theta - 2h \sin \theta \cos \theta) \\ &\quad - \{h(\cos^2 \theta - \sin^2 \theta) - (a - b) \sin \theta \cos \theta\}^2 \\ &= a^2 \cos^2 \theta \sin^2 \theta + ab \cos^4 \theta - 2ah \sin \theta \cos^3 \theta + ba \sin^4 \theta \\ &\quad + b^2 \sin^2 \theta \cos^2 \theta - 2bh \sin^2 \theta \cos \theta + 2ha \sin^3 \theta \cos \theta \\ &\quad + 2hb \sin \theta \cos^3 \theta - 4h^2 \sin^2 \theta \cos^2 \theta \\ &\quad - \{h^2(\cos^2 \theta - \sin^2 \theta)^2 + (a - b)^2 \sin^2 \theta \cos^2 \theta \\ &\quad - 2h(a - b)(\cos^2 \theta - \sin^2 \theta)(\sin \theta \cos \theta)\} \\ &= ab - h^2, \text{ on simplification.} \end{aligned}$$

Hence, the quantities $a + b$ and $ab - h^2$ of a second degree expression are **invariants** due to rotation of axes.

Observation: From a second degree equation in x and y ,

- (i) linear terms are eliminated by translation of axes and
- (ii) product term is eliminated by rotation of axes.

Summary of the chapter

Transformation of co-ordinates are mainly of two types: (i) Translation of axes and (ii) Rotation of axes.

Translation of axes means to alter the origin of co-ordinates without altering the directions of axes.

Rotation of axes means to change the directions of axes of co-ordinates, without changing the origin; both systems of co-ordinates being rectangular. *In case of translation of axes, following rules are to be followed:*

- (i) To get the values of the new co-ordinate (x', y') , subtract the co-ordinates of the new origin (h, k) from the original co-ordinates (x, y) respectively.
- (ii) To get the equation of any curve in terms of new co-ordinates (x', y') , substitute $x' + h$ and $y' + k$ for x and y respectively in the equation.

In case of rotation of axes, the following rules are to be followed:

- (i) If the co-ordinates (x, y) of any point referred to a given system of axes, its co-ordinates (x', y') referred to a system of new axes which have the same origin but rotated through an angle θ about the origin are obtained from the formulae.

$$x' = x \cos \theta + y \sin \theta \text{ and } y' = -x \sin \theta + y \cos \theta.$$

- (ii) If the equation of a curve referred to the original axes be given, the equation of the same curve referred to the new axes are obtained by substituting $(x' \cos \theta - y' \sin \theta)$ and $(x' \sin \theta + y' \cos \theta)$ for x and y respectively in the equation.

The main application of transformation of co-ordinates is to simplify an equation of second degree in x and y to the standard form of conic section which will be defined in chapter-VI.

Illustrative Examples

Ex.1. Transform the equation $x^2 - 3y^2 + 4x + 6y = 0$ to parallel axes through the point $(-2, 1)$.

Solution: Given equation is $x^2 - 3y^2 + 4x + 6y = 0$ (2.1)
Putting $x' - 2$ and $y' + 1$ for x and y respectively in the given equation (2.1), we have

$$(x' - 2)^2 - 3(y' + 1)^2 + 4(x' - 2) + 6(y' + 1) = 0$$

$$\text{or, } (x')^2 - 4x' - 3(y')^2 - 6y' + 4x' + 6y' - 1 = 0$$

$$\text{or, } (x')^2 - 3(y')^2 = 1$$

Dropping the dashes, we get $x^2 - 3y^2 = 1$

$\therefore$ required transformed equation is $x^2 - 3y^2 = 1$.

Ex.2. Transform the equation $x^2 - y^2 - 2\sqrt{2}x - 10\sqrt{2}y + 2 = 0$ when the axes are turned through $\frac{\pi}{4}$, the axes of co-ordinates being rectangular.

Solution:

Given equation is $x^2 - y^2 - 2\sqrt{2}x - 10\sqrt{2}y + 2 = 0$ (2.2)

To turn the axes through an angle $\frac{\pi}{4}$, we have to substitute

$$x' \cos \frac{\pi}{4} - y' \sin \frac{\pi}{4} \text{ and } x' \sin \frac{\pi}{4} + y' \cos \frac{\pi}{4}$$

$$\text{i.e., } \frac{1}{\sqrt{2}}(x' - y') \text{ and } \frac{1}{\sqrt{2}}(x' + y') \text{ for } x \text{ and } y \text{ respectively in}$$

equation (1) and doing so we get

$$\frac{1}{2}(x' - y')^2 - \frac{1}{2}(x' + y')^2 - 2(x' - y') - 10(x' + y') + 2 = 0$$

$$\text{or, } -2x'y' - 12x' - 8y' + 2 = 0$$

$$\text{or, } x'y' + 6x' + 4y' = 1$$

Hence the transformed equation is $xy + 6x + 4y = 1$, (dropping dashes).

Ex.3. Transform the equation

$$11x^2 + 24xy + 4y^2 - 20x - 40y - 5 = 0$$

to rectangular axes through the point $(2, -1)$ and inclined at an angle $\tan^{-1}\left(-\frac{4}{3}\right)$.

Solution: Given equation is

$$11x^2 + 24xy + 4y^2 - 20x - 40y - 5 = 0 \dots\dots\dots (2.3)$$

In this problem first change the origin to the point $(2, -1)$ and then rotate the axes through $\tan^{-1}\left(-\frac{4}{3}\right)$.

Putting $x' + 2$ and $y' - 1$ for x and y in equation (2.3), we get

$$11(x' + 2)^2 + 24(x' + 2)(y' - 1) + 4(y' - 1)^2 - 20(x' + 2) - 40(y' - 1) - 5 = 0$$

$$\text{or, } 11(x')^2 + 24x'y' + 4(y')^2 - 5 = 0$$

$$\text{or, } 11x^2 + 24xy + 4y^2 - 5 = 0 \dots\dots\dots (2.4)$$

(dropping the dashes)

Now to rotate the axes through $\tan^{-1}\left(-\frac{4}{3}\right)$, we take $\tan \theta = -\frac{4}{3}$ so

that $\sin \theta = -\frac{4}{5}$ and $\cos \theta = \frac{3}{5}$. Thus we have to substitute

$\frac{3}{5}x' + \frac{4}{5}y'$ and $-\frac{4}{5}x' + \frac{3}{5}y'$ for x and y respectively in equation

(2.4) which then becomes

$$11(3x' + 4y')^2 + 24(3x' + 4y')(-4x' + 3y') + 4(-4x' + 3y')^2 - 125 = 0$$

$$\text{or, } (x')^2 - 4(y')^2 + 1 = 0.$$

Dropping the dashes, we get $x^2 - 4y^2 + 1 = 0$ which is the required transformed equation.

i.e., if $\tan 2\theta = \frac{3}{4}$ i.e., $2\theta = \tan^{-1}\left(\frac{3}{4}\right)$.

Since $\tan 2\theta = \frac{3}{4}$, we have $\sin 2\theta = \frac{3}{5}$ and $\cos 2\theta = \frac{4}{5}$. If the value of θ thus obtained be substituted in (2.8) and the dashes be dropped, the equation on simplification reduces to $2x^2 - y^2 = 1$.

Thus if the equation (2.5) is transformed to rectangular axes through the point $(1, -1)$, inclined at an angle $\frac{1}{2}\tan^{-1}\frac{3}{4}$, it becomes $2x^2 - y^2 = 1$ which does not contain terms involving x, y and xy and hence $2x^2 - y^2 = 1$ is the required transformed equation.

Ex.5. Transform the equation $x^2 - y^2 = a^2$ when the axes are turned through 45° .

Solution: Given equation $x^2 - y^2 = a^2$ (2.9)

$$\text{Angle of rotation of axes} = 45^\circ = \frac{\pi}{4}$$

$$\text{Here, } x = x' \cos \theta - y' \sin \theta = x' \cos \frac{\pi}{4} - y' \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}(x' - y')$$

$$\text{and } y = x' \sin \theta + y' \cos \theta = x' \sin \frac{\pi}{4} + y' \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}(x' + y')$$

Putting these values of x and y in (2.9), we get

$$\frac{1}{2}(x' - y')^2 - \frac{1}{2}(x' + y')^2 = a^2$$

$$\text{or, } (x' - y')^2 - (x' + y')^2 = 2a^2$$

$$\text{or, } -4x'y' = 2a^2$$

$$\text{or, } 2x'y' + a^2 = 0$$

Dropping dashes, we get $2xy + a^2 = 0$ which is the required